

3D • 6DOF

Stereo 3D / 6DoF Manager

Complete Feature & Technical Reference

A one-click desktop manager for stereoscopic-3D and 6DOF head-tracking game mods

Game database	819 titles · 37 emulators
Managed mods	15 (3D drivers, VR export add-ons, 6DOF hubs, loaders)
Render APIs	DX7 / DX8 / DX9 / DX10 / DX11 / DX12 / Vulkan / OpenGL
Install engine	80 IPC handlers · 6 download strategies
Platform	Windows desktop (Electron) · offline-capable

This document describes what the application does, how it detects games, and the exact download & install method used for every mod. Community mods are the work of their respective authors (see Credits).

1 What This App Does

The **Stereo 3D / 6DoF Manager** turns a fragmented, manual process — hunting down 3D drivers, ReShade add-ons, per-game head-tracking mods, matching each to the right render API, and hand-editing config files — into a guided desktop workflow. It scans your drives, identifies each installed game, recommends the correct stereoscopic-3D pipeline, downloads the real mods from their official sources, installs them into the right folders with the correct proxy DLLs, and exposes their real settings in a clean editor.

Two things it manages

- **Stereoscopic 3D** — renders a separate image per eye for geometrically correct depth, output as Side-by-Side, Top-and-Bottom, interlaced, checkerboard, anaglyph, or full-resolution VR (to a VR viewer).
- **6DOF head tracking** — lets the in-game camera follow your head position and rotation (yaw / pitch / roll + lean) via community mods, reading pose over UDP from OpenTrack-compatible trackers.

Design principles

- **Real sources only** — every mod is downloaded from its official GitHub release / repository or shipped bundled; nothing is faked or reconstructed.
- **Correct by construction** — the right DLL name, bitness, subfolder layout, and config keys for each game's API, verified against each mod's own documentation.
- **One-click or manual** — a single button sets up a full pipeline, or you drive each step yourself.
- **Honest about limits** — when a source is unreachable or a config only appears after first launch, the app says so rather than pretending.

2 Glossary

Term	Meaning
Stereoscopic 3D	A separate image rendered for each eye so the scene has real depth on a 3D display or VR viewer.
6DOF	Six Degrees Of Freedom — head yaw, pitch, roll plus X/Y/Z lean driving the in-game camera.
Render API	The graphics interface a game uses: DirectX 7–12, Vulkan, or OpenGL. Determines which mod and proxy DLL apply.
Proxy DLL	A DLL named like a system graphics library (e.g. d3d9.dll, dxgi.dll) that the game loads, letting a mod hook in.
ReShade	A generic post-processing injector. Its Add-on build hosts stereo shaders and the VR-export add-ons.
SBS / TAB	Side-by-Side / Top-and-Bottom — two eye images packed into one frame. Half-res (squished) or full-res.
Katanga / VRScreenCap	VR viewer apps that display a full-res SBS frame shared to them by the VR-export add-ons.
Convergence	The virtual distance that sits on the screen plane. Nearer objects pop out; farther ones recede.
Separation	The 3D strength — how far apart the two eye viewpoints are (your real inter-pupillary distance).
ASI / BepInEx / etc.	Mod loaders that inject a plugin into a game (ASI Loader, BepInEx, REFramework, OWML).
Core	A shared, cached download of a mod that multiple games link against (vs. a per-game build).

3 Quick Start

① Scan

Click **Scan all drives**. The app reads each game's .exe header for its render API and 32/64-bit, matches the folder/exe name against the 819-title database, and organizes your library.

② Pick a game

Select a game. Its detected API and bitness appear under the title in a yellow selector you can override.

③ One-click setup

Choose an output format and press the setup button. The app picks the best 3D pipeline for the game's API, downloads it, installs it with the correct proxy DLL, and writes a working config.

④ Tune

Open **Config** to adjust the key 3D settings (separation, convergence, output) — pinned at the top, with everything else folded into an Advanced section.

⑤ 6DOF (optional)

On a game's card, install a head-tracking mod (auto-matched), or use the manual picker to choose any BerZerker or itsloopyo mod. Point your tracker at UDP 127.0.0.1:4242.

***Note:** Config files sometimes only exist after the game's first launch (loaders like BepInEx generate them on first run) — the app shows sensible defaults until then.*

4 The 3D Pipelines & Mods

The manager knows fifteen mods across five roles: stereoscopic-3D drivers, ReShade & its VR-export add-ons, per-game 6DOF hubs, and the loaders some mods need. It recommends a pipeline from the game's API.

Stereoscopic 3D drivers

Mod	Best for	How it outputs 3D
geo-11	DX11	Geometry-based stereo (3DMigoto lineage). Own direct_mode output incl. native full-res KatangaVR.
Geo3D (Flugan)	DX11 / DX12 / DX10	Injects real stereo; 3DToElse.fx converts to SBS/TAB/interlaced; GeoVrExport for full-res VR.
SuperDepth3D	Any API (depth-based)	Depth-map reprojection via ReShade. Built-in Stereoscopic_Mode; SuperVrExport for full-res VR.
wiz3D	DX9 & legacy	iZ3D-based proxy. Built-in OutputMethod DLLs: interlaced, SBS, DLP, anaglyph, shutter, SR-weave.

ReShade & VR export add-ons

Component	Role
ReShade (Add-on build)	Hosts the stereo shaders (SuperDepth3D.fx / 3DToElse.fx) and the VR-export add-ons.
SuperVrExport	Locked to SuperDepth3D. Copies the full-res SBS DoubleTex straight to KatangaVR / VRScreenCap.
GeoVrExport	Locked to Geo3D. Shares 3DToElse's texTOT to KatangaVR. Ships geod3d9.dll for the native D3D9 fast path.
Osiris VR Viewer	The VR display app the export add-ons feed.

6DOF head-tracking hubs

Source	What it provides
BerZerker hub	One release per game (43+ games). Some ship combined 3D + 6DOF all-in-one builds. Loaded via ASI.
itsloopyo	Per-game repos (33+ mods). Ships -nexus / -installer / -thunderstore builds; the app uses the -nexus files.
Loaders	BepInEx, Ultimate ASI Loader, REFramework — installed automatically when a mod needs one.

5 Output Formats

The one-click output picker adapts to the selected mod, because each mod produces 3D differently:

Mod	Built-in outputs	Full-res VR
geo-11	SBS, TAB, interlaced, checkerboard, 3D-Vision Direct, native KatangaVR	Own katanga_vr direct_mode (no add-on)
wiz3D	Interlaced, SBS, checkerboard/DLP, anaglyph, shutter, SR-weave	Built-in (SBS)
Geo3D	SBS, TAB, line/column interlaced, checkerboard, anaglyph (via 3DToElse.fx)	+ GeoVrExport add-on

Mod	Built-in outputs	Full-res VR
SuperDepth3D	SBS, TAB, line/column interlaced, checkerboard, quad, frame-seq	+ SuperVrExport add-on

Note: Half-res SBS/TAB packs both eyes into one frame (each eye is squished). Full-res VR sends a double-width frame to a VR viewer so each eye keeps full resolution.

6 Game Detection & Identification

How a game is identified

- **.exe header read** — the PE header gives the render API family and 32/64-bit without launching the game.
- **Name matching** — the folder and executable names are normalized and matched against the 819-title database, with smart collision handling (a remaster won't hijack the original, and vice-versa).
- **Manual override** — a yellow API/bitness selector under the game title lets you correct any detection; the choice persists across rescans and drives which mod & proxy the app uses.

Database coverage

The bundled database holds **819 games** plus **37 emulators**, each tagged with its engine and render API order (best → legacy). Representative API coverage:

API	Games	API	Games
DirectX 11	533	OpenGL	78
DirectX 12	202	Vulkan	60
DirectX 9	164	DirectX 10	10
DirectX 8 / 7	4	Emulators	37

Top engines represented include Unreal Engine 3/4/5, Unity, RE Engine, MT Framework, Frostbite, AnvilNext, Source, the FromSoftware engine, and many custom engines — so API/engine recommendations are accurate across the modern catalogue and legacy DX9 titles alike.

6DOF name matching

Head-tracking mods are matched to a game by *understanding its name*, not by an exact string — using acronym expansion and roman↔arabic numeral handling. "Batman Arkham Knight" matches "Batman: Arkham Knight"; "AC Unity" matches "Assassin's Creed Unity"; unrelated titles correctly get no match. If a match is wrong or missing, the manual picker lets you choose any mod from either catalogue.

7 Download & Install Methods

Every mod resolves through one of six strategies. The app caches shared cores, falls back to a bundled copy when a source is unreachable, and records every file it writes so uninstall is exact.

Strategy	Used for	How it works
github-release	SuperDepth3D core wrappers, wiz3D, Geo3D, VR-export add-ons, itsloopyo, Osiris	Reads the repo's latest release, picks the asset matching the game's API/bitness, extracts it.
release-hub	BerZerker 6DOF hub	One repo, one release per game; matches this game's tag/asset (incl. combined 3D+6DOF).
github-repo	ReShade shader repos	Downloads a branch zipball and merges Shaders/Textures.
archive-url	geo-11 (HelixMod), ReShade setup	Downloads a specific archive/installer from the official host.
bundled	Legacy Geo3D, VR-export add-ons + geod3d9.dll	Ships inside the app; installs offline with no download.
website	Loaders (guides)	Opens the official page with exact setup steps (no silent binary drop).

Install layout, per API

- **DX10 / DX11 / DX12** — ReShade loads as `dxgi.dll`; the VR-export add-on (`.addon32/` `.addon64`) sits next to it with no subfolder.
- **DX9 (native fast path)** — the VR-export add-on ships `geod3d9.dll`; see §8.
- **geo-11** — the loadable DLLs (`d3d11.dll`, `nvapi`, `d3dx*.ini`) land next to the exe, matched to bitness.
- **6DOF** — placed at the ASI/loader root the mod expects (BepInEx plugin, ASI next to exe, etc.).

Note: Uninstall removes only files the app itself installed (tracked in a per-game manifest), keeping game.exe and leaving a .bak of any config it touched.

8 The D3D9 Full-Res VR Proxy

Direct3D 9 games create a *plain* device that cannot share GPU textures across processes, so the VR-export add-ons would fall back to a slow per-frame CPU copy. To stay fully native, GeoVrExport ships a small proxy — `geod3d9.dll` — that upgrades the game to an extended (D3D9Ex) device so the fast GPU-shared path works. **This applies to both SuperDepth3D and Geo3D on DX9 games.**

Optional & manual — one button

On a DX9 game the manual install section shows an OPTIONAL button to set this up. When clicked, the manager builds the exact layout the add-on's documentation specifies — with the proxy **renamed to d3d9.dll** as the only DirectX DLL beside the executable, and ReShade moved into a subfolder it chainloads:

```
<game>\DAOrigins.exe
<game>\d3d9.dll ← geod3d9 proxy (renamed; the ONLY DirectX dll here)
<game>\ReShade\d3d9.dll ← ReShade itself (keeps the d3d9.dll name)
<game>\ReShade\GeoVrExport.addon32 · Geo3D.addon32
<game>\ReShade\ReShade.ini · ReShadePreset.ini
```

The load chain becomes: `game.exe` → `d3d9.dll` (proxy) → `ReShade\d3d9.dll` → system `d3d9`. ReShade's shader search paths are rewritten to point up one level so it still finds the shaders in the game root. Everything is tracked, so uninstall cleanly removes the proxy and restores the layout.

Preset arming (SuperDepth3D)

For SuperVrExport the app also seeds the exact values the add-on forces at startup — `Stereoscopic_Mode = 0` (Side-by-Side), `DoubleBuffer_Mode = 1`, and the `EX_DLP_FS_Mode=1`, `DoubleBuffer_Mode=1` preprocessor defines — so the full-res SBS → DoubleTex → KatangaVR path is armed and visible before first launch.

***Note:** Separately, legacy DX7/8/9/10 games have a built-in **Install dgVoodoo2** button that downloads the official package and places the correct DLL for the game's API + bitness, converting it to DX11 so `geo-11` / `ReShade` / `Geo3D` can run. `dgVoodoo2` is freeware by Dege, fetched unmodified from its source.*

9 Config Editor

The manager reads each mod's *real* config file from disk — even when it's named differently — and presents it with the settings that matter for 3D pinned at the top, and everything else folded into a collapsible Advanced section. Edits write straight back to the file (with a `.bak`), and are preserved across mod updates.

Mod	Pinned key 3D settings	Folded
geo-11	output mode, separation, convergence, auto-convergence, FPS monitor, force stereo	~168 advanced
wiz3D	output method, StereoBase, ZPS, separation scale, swap eyes, auto-focus	~90 advanced
SuperDepth3D	Stereoscopic_Mode (output), depth, ZPD/convergence, eye-swap, perspective, IPD	ZPD tuning, depth buffer, comfort, performance
Geo3D	separation, convergence, screen size, depth	per-fix values
ReShade	preset path, VR-arming defines, performance mode, depth access	all other keys
6DOF (combined)	enabled, output mode, separation mm, convergence distance, swap eyes	FOV, sensitivity, position, hotkeys

Values are the real defaults from each mod's own files (e.g. SuperDepth3D's `Zero_Parallax_Distance` default 0.025, `Depth_Adjustment` 50), and duplicate keys from overlapping files are de-duplicated so each setting

appears once.

10 Feature Summary

Feature	What it does
Drive scan	Finds installed games across drives, reads API + bitness from the .exe.
One-click setup	Picks & installs the best 3D pipeline for the game's API, writes a working config.
Manual install	Install any single mod yourself, with the correct proxy & layout applied.
Manual API / bit override	Yellow selector under the game title; drives mod choice; persists across rescans.
Dynamic output picker	Output list adapts to the selected mod's real formats, incl. VR add-ons.
6DOF auto-match	Understands the game name and matches a BerZerker or itsloopyo mod.
6DOF manual picker	Choose ANY mod & version from either catalogue, incl. combined 3D+6DOF builds.
Streamlined config editor	Key 3D settings pinned; the rest folded; edits saved with .bak; kept across updates.
Community shader fixes	Links geo-11 per-game 3D fixes from the Helix Mod list.
Uninstall	Removes only tracked files (per-game manifest), keeps game.exe.
Periodic update check	Background re-check for new mod releases (visibility-gated).
Backup & restore	Export/import all settings, library & profiles to a single JSON file.
Offline capable	Bundled cores + fallbacks install with no download when a source is blocked.

11 Technical Notes

- **Architecture** — Electron desktop app. A sandboxed renderer (the UI) talks to a privileged main process through **80 IPC handlers**; a preload bridge exposes a typed API. Install/scan logic lives in dedicated modules (scanner, installer, config, mods, gamedb, store).
- **Proxy DLL selection** — chosen per game API & bitness with clash-avoidance so two mods never fight over the same slot (DX10-12 → dxgi.dll, DX9 → d3d9.dll, OpenGL → opengl32.dll, Vulkan → setup-tool layer).
- **Config engine** — an INI/preset/XML parser that merges duplicate sections, de-duplicates keys, writes back only changed values, and always leaves a .bak.
- **Manifests** — each game keeps a manifest of installed mods, their versions, and every file written, so updates preserve tuning and uninstall is exact.
- **Resilience** — GitHub rate-limit tolerance, cached-core reuse, and bundled fallbacks keep installs working offline; the UI states clearly when a source can't be reached.

12 Credits & Licensing

This manager is an installer/orchestrator. The 3D and 6DOF mods it installs are the work of their respective authors and are downloaded from their official sources. Please support and credit them.

Project	Author	Role
SuperDepth3D / Depth3D	BlueSkyDefender	Depth-based stereoscopic 3D shader + VR
geo-11 / HelixMod	bo3b, DHR, Flugan & contributors	DX11 geometry 3D driver + per-game fixes
Geo3D	Flugan	D3D11 geometry stereo + 3DToElse
Super-VRExport / GeoVRExport	BerZerker96 (from artumino's ReshadeVRExport)	Full-res SBS export add-ons + geod3d9 proxy

Project	Author	Role
Katanga VR	bo3b	VR display app fed by the export add-ons
ReShade	crossire	Post-processing injector + add-on API
wiz3D	effcol	iZ3D-based legacy stereo wrapper
6DOF Hub	BerZerker96	Per-game 6DOF head-tracking mods
6DOF mods	itsloopyo	Per-game head-tracking mods
dgVoodoo2	Dege	DX8/9/10 → DX11 wrapper (DX9 fallback)
Loaders	BepInEx, ThirteenAG, praydog	BepInEx, Ultimate ASI Loader, REFramework

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An orchestrator for community stereoscopic-3D and 6DOF head-tracking mods. Not affiliated with or endorsed by the individual mod authors.